

Jacob Mattie

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Robotics and automation enthusiast, .

Programming

Bash Scripting
C++
HTML5 / SCSS
Linux CLI
MATLAB / Octave
Python
SQL

Connectivity

Bluetooth
Ethernet
HTTP
JSON
REST APIs
TCP/IP
WiFi

Middleware

Docker
EPICS
Gunicorn
journalctl
systemd
Linux
Redis

Testing/Data

CI/CD
ETL Pipelines
Root Cause Analysis
Test Design

BSc. Mathematics, 2022, Simon Fraser University

Work Experience

Investment Analyst — Data Architecture & Backend Systems

Aligned Capital Partners / River Wealth

Vancouver, BC

11/2024 - Ongoing

Q: What data exists that might let us predict stock growth and market behaviour?

- Engineered **data ingestion** workflows across multiple transport mechanisms (HTTP requests, file-based feeds), implementing stateful retries, reconciliation, and consistency checks to mitigate upstream connectivity failures.
- Fully provisioned and maintained **production Linux infrastructure** from bare-metal deployment through service orchestration (systemd, Nginx, Gunicorn), including failure recovery from system interruptions.
- Built Python-based reporting systems that **automated full-day monitoring workflows** with on-demand data surfacing for cross-departmental staff, improving execution speed and operational consistency.

Test Technologist

IBC Technologies

Burnaby, BC

6/2022 – 9/2023

Q: How can we safely optimize heat transfer from a boiler's combustion chamber?

- Automated hardware validation processes by leveraging IoT ESP32 microcontrollers and PLC systems, streaming test data into **cloud-connected repositories** for analysis and report generation.
- Led certification and **validation of hardware/software systems** for boilers under Canadian and U.S. regulations, triaging issues and coordinating cross-team root-cause resolution to support two product releases.
- Built an embedded **ESP32 control system** for a boiler lift assembly station, integrating GPIO laser obstruction sensors to improve technician input handling, increasing throughput by 20% and enhancing operator safety.
- Built high-throughput data analysis workflows in MATLAB for multi-million-point datasets, **reducing validation time by 85%**.

IT Technician (Internship)

BC Cancer Research Centre (BCCRC)

Vancouver, BC

09/2018 – 01/2019

Q: How much digital freedom can we grant researchers without compromising network security?

- Managed user access controls and permissions within a segregated research network, enforcing data isolation and **cybersecurity** policies across multiple user groups.
- Managed, diagnosed, and resolved **TCP/IP**-based network connectivity issues across a segmented domain environment, including endpoint access control and inter-device communication issues.
- Assisted in network recovery during facility-wide outage caused by a malware incident, supporting administrators in restoring connectivity and validating **system integrity** across endpoints.

Volunteering

Automation Technologist

TRIUMF - Canada's Particle Accelerator

Vancouver, British Columbia

05/2025 - Ongoing

Q: What modern technological advances have resolved longstanding engineering problems?

- Contributed the **technical foundations** of 3 scientific publications (in preparation) on distributed control systems in nuclear physics laboratories.
- Developed a containerized EPICS **deployment pipeline** (Docker + Bash), leading a modernization initiative for legacy middleware operating over \$1.5B in infrastructure.
- Coded a Raspberry Pi-based stepper motor control system, merging hardware wiring, driver design, and Python interface through a transistor array for **automated and repeatable** experimental control.